



KAOUTHAR KABIRI

PhD Student in AI For HealthCare

CONTACT



+212627002598



kaw.kabiri@gmail.com



Rabat , Morocco



<https://kawthar-tor.github.io/portfolio/projects>

OVERVIEW

I am Kaouthar Kabiri, a PhD student in Artificial Intelligence for Healthcare and Clinical Decision Support.

I have professional experience as a software engineer at Capgemini, working within an international team on large-scale projects for a French client.

My academic and professional background has strengthened my analytical thinking, problem-solving skills, and motivation to contribute to impactful research at the intersection of AI and healthcare.

TECHNICAL SKILLS

Programming Languages : Python, Java, C/C++

Machine Learning & Deep Learning : PyTorch, Keras/TensorFlow, CNN, RNN, Model Training, Model Evaluation

Natural Language Processing : Hugging Face Transformers, NLTK, spaCY

Data Analysis And Processing : Pandas, Numpy

Research & Scientific Tools : Latex, VosViewer

Computer Vision & Geometric Processing : OpenCV, scikit-image

Databases & Search Engine : PostgreSQL, SQL, ElasticSearch

SOFT SKILLS

Scientific writing, critical reading of research literature, analytical reasoning, interdisciplinary collaboration, research autonomy.

EDUCATION

PhD Student in Artificial Intelligence For Healthcare

National School of Applied Sciences (ENSA KENITRA) - UNIV. IBN TOFAIL

December 2025 - ongoing (early stage)

Thesis Topic: Explainable and Trustworthy Artificial Intelligence for Clinical Decision Support Systems

Initial phase of doctoral studies :

- Literature review and state-of-the-art analysis in AI for Healthcare
- Exploration of explainability methods for medical AI systems

Computer Science Engineering Program

2019 - 2024

ENSA MARRAKECH - UNIV.CADI AYYAD

A five-year post-baccalaureate program taught in French, leading to an engineering degree in Computer Science.

The curriculum includes two years of preparatory studies followed by three years of specialized engineering training.

Degree equivalence : Master of Science (MSc)

GPA : 14 / 20 (equivalent to approximately 3.3 / 4.0)

International Baccalaureate (French Curriculum)

2018 - 2019

Specialization: Mathematical Sciences – Option A (Life and Earth Science)

Distinction: Mention Bien (Good)

PROFESSIONAL EXPERIENCE

Software Engineer - Java | Spring | Spring Batch | Apache Camel | PostgreSQL | PL/pgSql | JEE | Junit | SonarQube

August 2024 - December 2025

Capgemini TS Rabat



- Analysis and resolution of application issues (log analysis, bug reproduction, and root-cause diagnosis).
- Fixing defects in a JEE/Spring application, in compliance with development standards and code quality best practices, including the implementation of unit tests.

Software Engineer - End-of-Studies Internship

March 2024 - August 2024

Capgemini TS Rabat



- Addition of CFT configurations and implementation of Apache Camel to automate the transfer of files generated by batch processing to remote servers.
- Execution of unit tests using JUnit and deployment to the production environment through CI/CD pipelines.
- Debugging and defect resolution.

PROJECT

MRI to 3D Mesh Reconstruction

Completed

- Description : Python-based medical imaging pipeline to convert a series of 2D MRI slices into a 3D surface mesh. The system processes DICOM images, reconstructs volumetric data, and applies surface extraction techniques to generate accurate 3D anatomical models.
- Technologies : [#Python](#) [#3D Reconstruction](#) [#Computer Vision](#) [#Medical Imaging](#)
- Github Repository : <https://github.com/kawthar-tor/MRI-To-3D-Mesh>

Bone Fracture Detection from X-ray Images

Completed

- Description : This project focuses on developing an artificial intelligence system capable of detecting bone fractures from X-ray medical images. A deep learning model based on convolutional neural networks (CNN) was trained to classify radiographic images into fractured and non-fractured cases.
- Technologies : [#Python](#) [#PyTorch](#) [#Deep Learning](#) [#CNN](#) [#Medical Imaging](#)

Children Pneumonia Detection from Chest X-ray Images

Completed

- Description : This project focuses on building an intelligent medical imaging system to automatically detect pneumonia in children from chest X-ray images. A deep learning pipeline based on convolutional neural networks (CNN) was designed to classify radiographic images as either normal or pneumonia cases.
- Technologies : [#Python](#) [#Convolutional Neural Networks](#) [#Computer Vision](#)

Brain Tumour Detection and Classification

Completed

- Description : This project focuses on developing an intelligent medical imaging system capable of detecting and classifying brain tumours from MRI scans. Using deep learning techniques, the application analyzes brain MRI images to automatically determine whether a tumour is present.
- Technologies : [#Medical Imaging](#) [#FastAI](#) [#Python](#) [#Deep Learning](#)

Liver Tumor Segmentation and Volume Estimation

Completed

- Description : This project focuses on the automatic segmentation of liver tumors from 3D medical scans (CT/MRI images) using advanced medical image processing techniques. The system identifies tumor regions within volumetric liver data.
- Technologies : [#3D Segmentation](#) [#Medical Image Processing](#) [#Python](#) [#CT/MRI Imaging](#)
- Github Repository : <https://github.com/kawthar-tor/LiverTumor3DSegmentation>

Apache Log Analyzer

Completed

- Description : This is a desktop application developed in Java, using the JavaFX package to integrate interface components. This application allows real-time visualization of the results of data analysis recorded in the logs of a web server.
- Technologies : [#MultiThreading](#) [#Java](#) [#Dashboard](#) [#Regex](#) [#JavaFX](#)